Jack Benarroch Jedlicki

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Education

2025–Present **PhD in Computer Science**, *Harvard University*
Working with Prof. Heng Yang and Prof. Yilun Du. GPA: 4.00/4.00.
Funded by the "La Caixa" Foundation Fellowship and Harvard SEAS Prize Fellowship.
Coursework: Robotic Manipulation (MIT), Optimal Transport for Machine Learning, Foundations of Reinforcement Learning, Advanced Topics in Computer Vision, Advanced Topics in Modern Machine Learning.

2022–2023 **Visiting Student**, *UC Berkeley*
Fully funded academic year. GPA: 4.00/4.00.

2019–2024 **BSc in Mathematics & BSc in Physics**, *University of Barcelona*
Selective dual-degree program (20 students admitted annually). GPA: 9.5/10.
Ranked 2nd in the dual-degree cohort; 3rd/141 in Mathematics and 3rd/198 in Physics.

Research Experience

2025–Present **Graduate Researcher**, *Harvard University*, Cambridge, MA
Computational Robotics Lab: embodied AI agents, memory, robot learning, and perception.
- **Current research:** Developing embodied agents and a benchmark for long-horizon mobile manipulation and spatiotemporal memory; designing an end-to-end agentic system that integrates perception, action, memory, and new skill acquisition.
- **Frontier VLMs as embodied agents:** Designed an embodied agent using a frontier VLM as an end-to-end robot policy for memory-dependent manipulation through a simple harness. Surpassed human performance and the best leaderboard score on the RoboMME memory-conditioned manipulation benchmark (first author).
- **OCC4M:** Developed object-centric 4D memory for spatiotemporal reasoning and long-horizon manipulation, evaluated in simulation and on a real Franka arm. Accepted at NeurIPS 2026 Embodied Spatial Reasoning Workshop and as an **IROS 2026 workshop Spotlight** (first author).
- **Ledger:** Coauthored research on an object ledger for memory-augmented vision-language-action models, accepted at NeurIPS 2026 Embodied Spatial Reasoning Workshop (joint first author).

2024–2025 **Engineering Intern, AI Department**, *Nostrum Biodiscovery*, Barcelona, Spain
Developed deep learning methods for RNA structure prediction using RNA language models.

2024 **Research Intern**, *Barcelona Supercomputing Center*, Barcelona, Spain
Developed tensor-network methods and a software library for data compression and quantum-inspired physics simulations. Physics bachelor's thesis ranked among the top three.

2023–2024 **Research Intern**, *Computer Vision Center*, Barcelona, Spain
Advised by Dr. Sergio Escalera and Dr. Carles Casacuberta. Developed topological regularization for generative models, implemented the *topogen* library, and proved new differentiability results for topological losses. Top-ranked Mathematics bachelor's thesis; awarded Elena Maseras Research Fellowship.

2023 **Research Intern**, *UC Berkeley, Alaa Lab*, Berkeley, CA
Developed cardiovascular digital twins using computer vision and physics-informed neural networks. Resulting publication at **NeurIPS 2024** (joint first author).

2022 **Research Intern**, *MIT, Auto-ID Lab*, Cambridge, MA
Takeda-sponsored research on metal-particle detection for vaccine inspection. Resulting publication in **IEEE Sensors Journal** (first author).

Publications & Presentations

2026 **J. B. Jedlicki**, A. Sethi, H. Yang. "Can a Frontier VLM Serve as a Robot Policy for Memory-Dependent Manipulation?"
arXiv preprint. arXiv:2610.17350.

2026 T. Dieudonné*, **J. B. Jedlicki***, H. Yang. "Where Memory Belongs: Ledger, an Object Ledger for Memory-Augmented VLAs."
The 2nd Workshop on Embodied Spatial Reasoning, NeurIPS 2026. arXiv:2609.34554. (*Joint first authors.)

2026 **J. B. Jedlicki**, T. Dieudonné, H. Yang. "OCC4M: Object-Centric 4D Memory for Spatiotemporal Reasoning in Long-Horizon Manipulation."
The 2nd Workshop on Embodied Spatial Reasoning, NeurIPS 2026; *IEEE IROS 2026 Hybrid Architectures for Embodied Autonomy Workshop* (**Spotlight**). arXiv:2609.28798.

2026 E. Hahami, Y. Zimmermann, R. Zhou, **J. B. Jedlicki**. "A Unifying Lens on Reward Uncertainty in RLHF."
arXiv preprint. arXiv:2606.09073.

2024 K. Kuang*, F. Dean*, **J. B. Jedlicki***, D. Ouyang, A. Philippakis, D. Sontag, A. Alaa. "Med-Real2Sim: Non-Invasive Medical Digital Twins using Physics-Informed Self-Supervised Learning."
NeurIPS 2024. (*Joint first authors.)

2024 **J. B. Jedlicki**. "TopoGEN: Topology-Informed Generative Models."
Mathematics Bachelor's Thesis. Also presented at Computer Vision Center Talent Day.

2024 **J. B. Jedlicki**. "Tensor Networks for Quantum-Inspired Simulations."
Physics Bachelor's Thesis.

2023 **J. B. Jedlicki**, D. Tellbach, B. Subirana. "Metal Particle Detection Methods and Their Use for Freeze-Dried Vaccine Inspection: A Review."
IEEE Sensors Journal 2023.

Selected Projects

2025 **Diffusion Policy: Is Scaling All You Need?** *MIT 6.4212 final project*.
Studied generalization and scaling of diffusion policies under distribution shifts, including novel object configurations and obstacle-rich environments.

2025 **Are OT-Aligned Flows What You Need?** *CS 2840 final project*.
Compared diffusion models, flow matching, and rectified flows for image and robotic action generation; investigated one-step inference for robot actions.

Selected Fellowships & Awards

2025 **Harvard SEAS Prize Fellowship**: two years of PhD funding.

2024 **"La Caixa" Foundation Fellowship**: two years of full PhD funding; awarded annually to 35 exceptional STEM students from Spain and Portugal.

2024 **NYU Henry M. MacCracken Fellowship** (declined): five-year fellowship providing full funding for the PhD in Applied Mathematics.

2023 **Elena Maseras Research Fellowship**: one of eight recipients in Spain.

2022 **UB-UCEAP Scholarship**: one of five University of Barcelona recipients.

Teaching & Skills

2026–Present **Teaching Fellow, Harvard University**: AM 10, Computing with Python for Scientists and Engineers.

Technical Python, C++, PyTorch, Linux, Git; ManiSkill, MuJoCo; diffusion policies; VLA fine-tuning (e.g., $\pi_{0.5}$) and real-world deployment (e.g., Franka, Sharpa hands, TeleAvatar 2).

Languages Spanish, French, Catalan (native); English (fluent; IELTS 8.0).